Decoder ladders and predecessor blockers at the record atom 1113

27 August 2026

Let

$$x_{t+1} = U_+(x_t) := \frac{3x_t + 1}{2^{a_t}}, \qquad a_t = v_2(3x_t + 1), \tag{1}$$

be an accelerated orbit of positive odd integers. On an interior deficit-four record tail, write

$$A := 1, \qquad B := 12, \qquad C := 1113, \qquad D := 1122 \tag{2}$$

for the four record atoms, and put $b(x) := v_3(x+1)$ at a record state. Their decoder transport is

$$A : b \mapsto b + 1, \qquad B : b \mapsto 0, \qquad C : b \mapsto 1, \qquad D : b \mapsto 0. \tag{3}$$

For a normalized unbounded ray, let Q denote the coefficient at a specified state and let $\mathcal{K}$ be its stationary height. The synchronized companion satisfies

$$2^{a_t} e_{t+1} = 3e_t - 1, \qquad e_t > 1, \qquad x_t + e_t = \mathcal{K} Q_t. \tag{4}$$

Theorem 1 (the complete decoder ladder below a C atom)**.** *Suppose that a record transition starts at x, has atom $C = 1113$, and ends at y. Put*

$$b := v_3(x + 1). \tag{5}$$

For every integer s with $0 \le s \le b$, define

$$\boxed{z_s + 1 := \frac{2^{s-1}(x+1)}{3^s}.} \tag{6}$$

At $s = 0$, formula (6) means $z_0 = (x-1)/2$. Every z_s is a positive odd integer, its exact accelerated word to y is

$$\boxed{1^s 1121,} \tag{7}$$

and hence

$$U_+^{s+4}(z_s) = y. \tag{8}$$

If e_x is the synchronized companion value at x, put

$$\epsilon_s := 1 + \left(\frac{2}{3}\right)^s \frac{e_x - 1}{2}. \tag{9}$$

Then $\epsilon_s > 1$, it is transported to the original companion at y by the word in (7), and

$$\boxed{z_s + \epsilon_s = \frac{2^{s-1}}{3^s}(x + e_x).} \tag{10}$$

If x lies on the orbit of a globally least positive odd unbounded seed n, then, for every $0 \le s \le b$,

$$\boxed{x + 1 \ge 2(n+1)\left(\frac{3}{2}\right)^s.} \tag{11}$$

If Q is the coefficient at x, then

$$\boxed{Q > \frac{2(n+1)}{\mathcal{K}}\left(\frac{3}{2}\right)^s.} \tag{12}$$

In particular, (11)–(12) hold with $s = b$.

Proof. The exact C congruence and affine formula are

$$x \equiv 79 \pmod{128}, \qquad y = \frac{81x + 65}{64}. \tag{13}$$

Thus $v_2(x+1) = 4$. The state $z_0 = (x-1)/2$ has exact word 1121 and reaches y.

For $0 \le s \le b$, equation (6) gives

$$z_s + 1 = \frac{2^s(z_0 + 1)}{3^s}, \qquad v_2(z_s + 1) = s + 3. \tag{14}$$

Under an exponent-one step, the shifted state $X + 1$ is multiplied by $3/2$. Consequently the first s steps from z_s are exact exponent-one steps and end at z_0. Appending the exact word 1121 proves (7)–(8).

Backward transport of a companion through one exponent-one step is the map

$$E \longmapsto \frac{2E+1}{3}, \qquad \frac{2E+1}{3} - 1 = \frac{2}{3}(E-1). \tag{15}$$

At z_0, the companion value is $(e_x+1)/2$. Iterating (15) proves (9), and addition of the state and companion formulas gives (10).

Every z_s reaches the unbounded tail, so global leastness gives $z_s \geq n$. Equation (6) now proves (11). Since $x + e_x = \mathcal{K}Q$, equations (10), $z_s \geq n$, and $\epsilon_s > 1$ give

$$\frac{2^{s-1}}{3^s}\mathcal{K}Q = z_s + \epsilon_s > n+1,$$

which is (12). □

Corollary 2 (pullback through a one-atom run)**.** *Suppose that a record state p is followed by the atom block A^rC, where $r \geq 0$, and let y be its terminal state. Then*

$$\boxed{h := \frac{p-1}{2}} \tag{16}$$

is a positive odd integer with exact word 1^r1121 to y. If e_p is the synchronized companion at p, the alternative companion at h is $(e_p+1)/2$, and

$$h + \frac{e_p+1}{2} = \frac{p+e_p}{2}. \tag{17}$$

On a globally least unbounded ray, $p \geq 2n+1$.

Proof. Let x be the start of the terminal atom C. The r atoms A give

$$x+1 = \frac{3^r(p+1)}{2^r}, \qquad b(x) = b(p) + r \geq r. \tag{18}$$

Taking $s = r$ in (6) gives $z_r = (p-1)/2$. The assertions now follow from Theorem 1: indeed, forward companion transport through A^r gives $e_x - 1 = (3/2)^r(e_p-1)$, so (9) becomes $\epsilon_r = (e_p+1)/2$. □

Lemma 3 (least four-step predecessor of half a C endpoint)**.** *Suppose that a C atom starts at q and ends at p, and put*

$$h := \frac{p-1}{2}. \tag{19}$$

If h is odd, then every positive odd w satisfying $U_+^4(w) = h$ obeys

$$\boxed{w \geq 4q-1.} \tag{20}$$

Equality holds, and the unique exponent word attaining it is 1215.

Proof. The affine equation for C gives

$$p = \frac{81q + 65}{64}, \qquad h = \frac{81q + 1}{128}, \qquad h \equiv 50 \pmod{81}. \tag{21}$$

In particular, $q \equiv 79 \pmod{128}$ and $q \geq 79$.

Let $v = (c_1, c_2, c_3, c_4)$ be the exact word from w to h, let $B = c_1 + c_2 + c_3 + c_4$, and put

$$C(v) = 27 + 9\,2^{c_1} + 3\,2^{c_1+c_2} + 2^{c_1+c_2+c_3}. \tag{22}$$

Then

$$81w + C(v) = 2^B h. \tag{23}$$

Writing

$$s_1 = c_1, \qquad s_2 = c_1 + c_2, \qquad s_3 = c_1 + c_2 + c_3,$$

the integrality condition modulo 81 is

$$27 + 9\,2^{s_1} + 3\,2^{s_2} + 2^{s_3} \equiv 50\,2^B \pmod{81}, \qquad 1 \leq s_1 < s_2 < s_3 \leq B - 1. \tag{24}$$

Using the residues of 2^j modulo 81, the finite check for $4 \leq B \leq 9$ is

B	4	5	6	7	8	9
$50\,2^B \bmod 81$	71	61	41	1	2	4
# solutions of (24)	0	0	0	0	0	1.

(25)

For $B = 9$, the unique solution is $(s_1, s_2, s_3) = (1, 3, 4)$, hence

$$v = 1215, \qquad C(v) = 85. \tag{26}$$

Equations (21), (23), and (26) give

$$w = \frac{512h - 85}{81} = 4q - 1. \tag{27}$$

The five states for the proposed word are

$$4q - 1, \quad 6q - 1, \quad \frac{9q - 1}{2}, \quad \frac{27q - 1}{4}, \quad \frac{81q + 1}{128} = h.$$

The congruence $q \equiv 79 \pmod{128}$, together with the hypothesis that h is odd, shows that the extracted powers of two are exactly $1, 2, 1, 5$.

It remains to exclude a smaller predecessor with $B \geq 10$. Since every $c_i \geq 1$, equation (22) gives

$$C(v) \leq 27 + 9\,2^{B-3} + 3\,2^{B-2} + 2^{B-1} = 27 + \frac{19}{8}2^B. \tag{28}$$

As $h \geq 50$ and $B \geq 10$,

$$\begin{aligned}(2^B - 512)h - (C(v) - 85) \\ \geq 50(2^B - 512) - \left(\frac{19}{8}2^B - 58\right) > 0.\end{aligned} \tag{29}$$

Subtracting (27) from (23) now gives $w > 4q - 1$. This proves (20) and the equality statement. □

Theorem 4 (nontrivial record atoms block the half-seed pullback)**.** *Suppose that a nontrivial atom $P \in \{B, C, D\}$ ends at a record state p, and that p is followed by $A^r C$ for some $r \geq 0$. Let $h = (p-1)/2$ be the exact coalescing seed from Corollary 2.*

1. *If $P = B$ or $P = D$, then $3 \mid h$. Hence h has no accelerated positive odd predecessor.*
2. *If $P = C$ starts at q, then every four-step accelerated pullback of h is at least $4q - 1 > q$.*

Thus the half-seed construction propagates backward through the complete intervening run of atoms A, but across the preceding nontrivial atom it either terminates or expands beyond that atom's starting state.

Proof. Write $Y = p + 1$. If $P = B$, its shifted-state equation has the form

$$8Y = 9Y_0 + 4,$$

and if $P = D$, it has the form

$$64Y = 81Y_0 + 56.$$

In both cases $Y \equiv 2 \pmod 3$. Therefore

$$h = \frac{Y}{2} - 1 \equiv 0 \pmod 3.$$

A multiple of 3 has no accelerated odd predecessor, proving the first part. The second part is Lemma 3. □

Corollary 5 (localized remaining gate in the infinite-C branch)**.** *On an interior record tail with infinitely many atoms C, every local C half-coalescence has the complete ladder (6)–(12). Pulling it toward the fixed initial seed by the preceding atoms A is exact. Apart from at most the first C after the*

chosen tail origin, the first preceding atom in $\{B, C, D\}$ is a blocker in the sense of Theorem 4. Consequently, any global descent obtained from infinitely many C atoms must use an inverse branch other than the direct half-seed ladder across at least one nontrivial record atom.

Proof. Apply Corollary 2 to the maximal run A^r preceding each selected atom C, and then apply Theorem 4 to the preceding nontrivial atom. □